

FMCG Multi-Country Sales Dataset Analysis

(Data Profile)

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2.1 Dataset File Description

The dataset '*fmcg_sales_3years_1M_rows.csv*' is a highly realistic FMCG sales data generated for professional demand planning, forecasting, and supply chain analytics. The author/source of the dataset claims such:

- I. 1.1M+ sales records
- II. 15 countries, 60+ cities (with Lat/Lon coordinates)
- III. 350+ SKU
- IV. 3 full years of sales data
- V. Multiple sales channel (Modern trade, Traditional trade, Wholesale, E-commerce)
- VI. Promotional effects & Price elasticity
- VII. Seasonality patterns (weekly, monthly, yearly)
- VIII. ABC/XYZ segmentation
- IX. Lead time variability
- X. Stockouts & safety stock parameters

2.2 Fields Identity

The dataset contains 33 total fields. Here are some of the generic field names, types and their meanings are mentioned below:

Field Name	Data Type	Description	Category
date	Date	Daily timestamp (continuous 3 years)	Time (unique)
sku	Text	Product identifier	Product
category	Text	FMCG categories (Beverage, Snacks, ...)	Product
country	Text	Business countries	Geography
city	Text	Specific cities	Geography
latitude, longitude	Decimal	Geographic coordinates	Geography
channel	Text	Business types (MT, TT, WS, E-Com)	Distribution
units_sold	Whole num.	Daily units sold	Sales/Demand
list_price	Decimal	Base price	Pricing
promo_flag	Whole num.	Promotion status (1 – On, 0 – Off)	Promotion
Discount_pct	Decimal	Percent discount	Promotion
stock_out_flag	Whole num.	Stock condition (0- In stock, 1 – Out of stock)	Customer serv.
purchase_cost	Decimal	Procurement cost per unit	Finance
margin_pct	Decimal	Margin computation	Finance
stock_on_hand	Whole num.	Stocks available	Inventory
lead_time_days	Whole num.	Supplier lead time	Logistics
supplier_id	Whole num.	Supplier identity code	Sourcing

2.3 Data Quality

After understanding what each data means in a dataset, the next job of a data analyst is to perform Data Quality Check. We don't just blindly believe what the source/author claims, we have to do inspections and make sure the dataset is error free and perfectly structured to perform further analysis.

This is my findings on this specific dataset against authors description:

Parameters	Author/Source claimed	My findings
Total Records	1.1M+	Exact 1.1M
SKUs	350+	102
Countries	15	7
Cities	60+	9
Sales timeline	3 years	3 years (1-Jan-21 to 31-Dec-23)
Distinct product cat.	Not mentioned	5 categories
Sales channel	4 categories	4 categories
Num. of sales per date	Not mentioned	1004 ~1005 sales per dates

Records distributed in 7 countries:

Countries	Records
Germany	175200
Italy	350400
Poland	87600
France	87600
Spain	262800
Austria	87600
Netherlands	48800
Total	1.1M

List of data inconsistency:

Parameters	Quantity
Number of rows that have at least 1 Null/Blank value	0
Duplicate Rows	0
Data type errors (in suspected fields)	0%
Typo/mismatch errors (in suspected fields)	0
Data trimming required	0

All these outcomes prove that the dataset is well structured and meticulously inspected by the source to mitigate errors. (This is expected for a synthetic dataset, which is not usual in real world datasets. That's why we need to perform data cleanup)

2.4 Dataset Cleanup Methods

[1] Find out rows that have at least 1 Null/Blank value throughout the records (across all 33 columns):

- Step 1: Go to the **Add Column** tab and click **Custom Column**.
- Step 2: Name the column 'Null_Columns' and use the following code formula:
`=List.Contains(Record.FieldValues(_), null)`
 (This scans every cell in that specific row for a null value and returns TRUE or FALSE).
- Step 3: Filter this new column to only show TRUE.
- Step 4: Go to the **Transform** tab and click **Count Rows**. Note this number down to get your result on number of rows with blank/null entity.

Fix: Go back to the custom column step you created in Test 1 (Null_Columns). Filter that column to keep only FALSE Right-click the 'Null_Columns' column and select **Remove**

[2] Number of Duplicate rows based on unique field (in this dataset, 'dates' field is unique):

- **Step 1:** Right-click your main query in the left pane and select **Reference** (this creates a linked, lightweight copy so you don't mess up your original data).
- **Step 2:** In the new query, select only the **Sales Date** column. Right-click the header and select **Remove Other Columns**.
- **Step 3:** Go to the **Home** tab, click **Keep Rows > Keep Duplicates**.
- **Step 4:** Go to the **Transform** tab and click **Count Rows**. This will give you the exact number of non-unique sales date records. The difference between current row count and previous row counts express the number of duplicate rows present.

Fix: Remove this action from the right-side pane. Then right click on the unique field header (dates) and select 'Remove duplicates'

[3] Data type errors in suspected fields:

- Go to the **View** tab and check **Column Quality**. It will instantly show you an "Error" percentage bar right below the column header.

Fix: Select the column containing the data type errors. Go to the **Home** tab. Click **Remove Rows > Remove Errors** (if you want to delete those rows completely).

[4] Typo/Mismatch errors in suspected fields/columns (e.g., 'aple', 'apple' instead of 'Apple'):

- **Step 1:** Go to the **View** tab and check the **Column Profile** and **Column Distribution** boxes.
- **Step 2:** Click on your suspected text column header.
- **Step 3:** Look at the **Column Profile** pane at the bottom of your screen. It will show a list of **Value Distribution**.
- **Step 4:** Click the **Value** column header inside that profile pane to sort alphabetically. You will instantly see 'aple', 'apple', and ' ' grouped next to each other, along with the exact count of how many times each typo appears in your 1.1M rows.

Fix: Go to the **Transform** tab and click **Group By**. Select **Advanced** at the top. Set your grouping column (e.g., Product Name). Check the box at the bottom that says **Use fuzzy grouping**. Click **Fuzzy group options** to adjust how strict it is (a similarity threshold of 0.8 usually catches things like 'aple' and 'apple'). Set the **Operation** to All Rows. This creates a cleaned master table where the typos are automatically mapped to the most common, correct spelling.

[5] Trims/Spacing issues in suspected columns (e.g., 'PersonalCare' instead of 'Personal Care'):

- **Step 1:** Select your suspected column.
- **Step 2:** Go to the **Add Column** tab and click **Conditional Column**.
- **Step 3:** Set the rule: *If Column Contains* (type two spaces) or *If Column Contains* PersonalCare, then output 1, else 0.
- **Step 4:** Filter the new column to only show 1 and use **Transform > Count Rows** to see the volume of records requiring text cleanup.

Note: If you just want to see how many rows have accidental *leading or trailing* spaces, you can add a custom column with text length comparisons:

`Text.Length([Column]) = Text.Length(Text.Trim([Column]))`. Any row returning FALSE needs a trim.

Fix:

- **To fix Leading/Trailing Spaces:**

Select your text columns (hold Ctrl to select multiple text fields). Right-click any highlighted header and go to **Transform > Trim**. *(This instantly strips spaces from the very front and back of your words).*

- **To fix missing internal spaces (e.g., 'PersonalCare' → 'Personal Care'):**

Right-click the column header and select **Replace Values**. In **Value to Find**, type "PersonalCare". In **Replace With**, type "Personal Care". Click **OK**

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